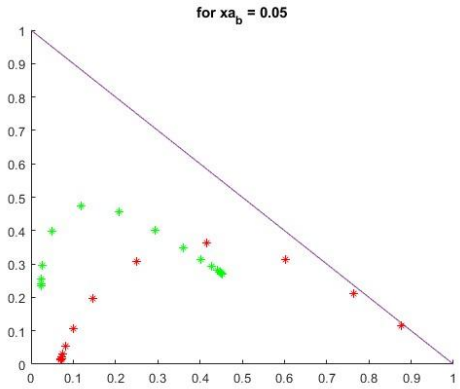
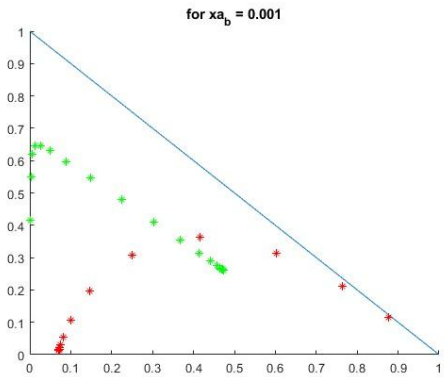
SHRIMI AGRAWAL

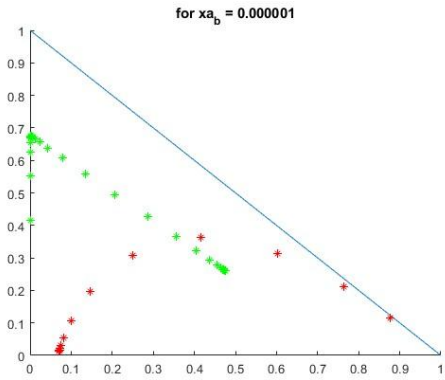
221036



A)



B)



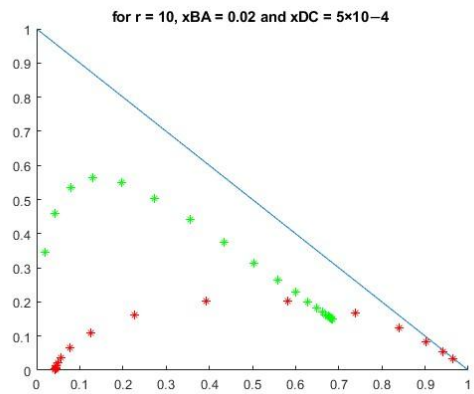
C)

For [αAC αBC] = [6.35 2.47], xF = [0.3 0.3 0.4], x _D_A = 0.95, x_D_B = 0.049 and r = 2.5,

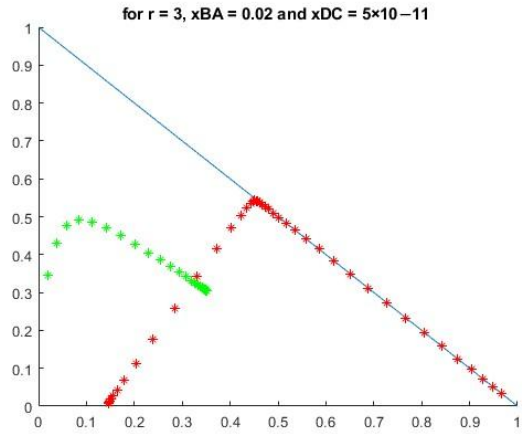
	no of plates
a) x B A = 0.05	5(R) + 8(S)
b) x B A = 0.001	5(R) + 11(S)
c) x B A = 0.000001	5(R) + 12(S)

We observe as we increase the molar ratio for stripping section stream the intermediate boiler effect becomes more apparent. As we go more closer towards pure binary mixture saddle point becomes more apparent.

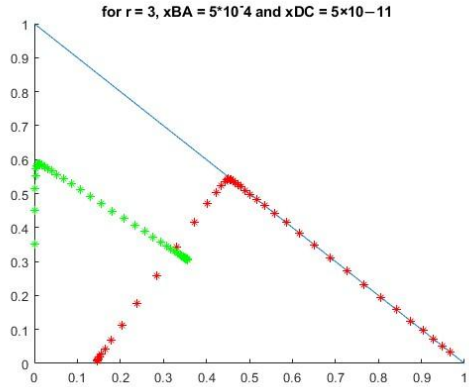
from graph both stripping section and rectifying section intersection point tell the ideal number of plates required .



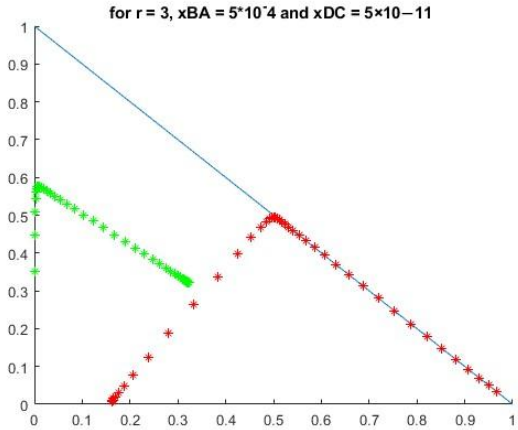
a



b



c



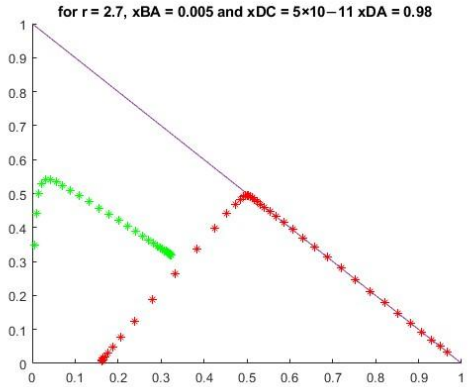
d

For [α AC αBC] = [3.25 1.90], xF = [0.3 0.25 0.45] and x D A = 0.98

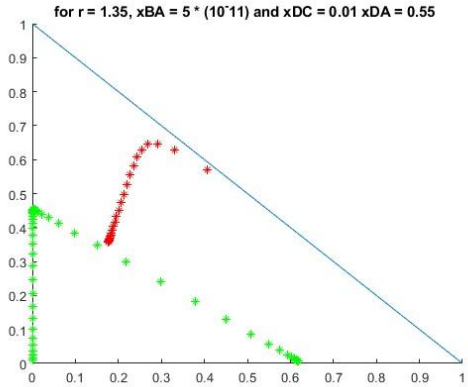
		No of trays required(ROUGH ESTIMATE)
a)	r = 10, x B A = 0.02 and x D C = 5×10−4	6(R) + 12(S)
b)	r = 3, x B A = 0.02 and x D C = 5×10−11	32(R) + 14(S)
c)	r = 3, x B A = 5×10−4 and x D C = 5×10−11	28(R) + 28(S)
d)	r = 2.7, x B A = 5×10−4 and x D C = 5×10−11	30(R) + 50(S)

here for d part we observe that rectifying and stripping column which indicate separation for this specific condition is not feasible even after utilising all 50 plates the separation is not observed.

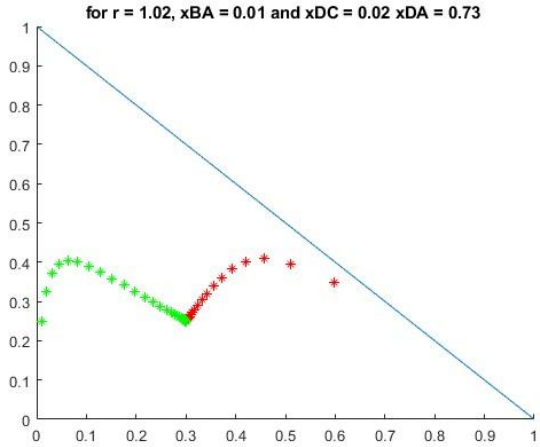
Here from a and b we see when we increase r the value for plates in rectifying column increases significantly



a)



b)



c)

	no of total plates
a) r=2.7, xD =[0.98 0.02 5×10−11] and xB =[0.005 0.350 0.645]	30 (R) + 50(S) (SINCE DO NOT INTERSECT)
b) r=1.35, xD =[0.55 0.44 0.01] and xB =[5×10−11 0.022 0.978]	37(S) + 50(R)
c) r=1.02, xD =[0.73 0.25 0.02] and xB =[0.01 0.25 0.74]	50(R and S)